

Avner May

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WORK EXPERIENCE

Together.ai

Staff Research Scientist, Efficient LLM inference

Developing and productionizing efficient inference algorithms, focusing on speculative decoding, reporting directly to CTO. Close collaboration with Tri Dao, Chief Scientist & Flash Attention creator. Example projects include Speculative Speculative Decoding, MagicDec, Sequoia.

New York, NY
Sep. 2023 - present

Google

Research Scientist, Speech Recognition

Worked on self-supervised learning for audio-visual speech recognition models. The goal was to use untranscribed videos of people speaking (e.g., audio + lip movements) to improve the performance of audio-visual speech recognition systems.

New York, NY
Oct. 2020 - Sep. 2023

Microsoft

Software Development Engineer

Designed and implemented features to facilitate the development of distributed applications.

New York, NY
Aug. 2009 - Aug. 2011

EDUCATION

Stanford University

Postdoctoral Scholar

Advisor: Christopher Ré

Stanford, CA
Jan. 2018 - Jul. 2020

Columbia University

MS/PhD in Computer Science

Dissertation: Kernel Approximation Methods for Speech Recognition

Advisor: Michael Collins

Honors: Recipient of the Department Chair's Distinguished Fellowship

New York, NY
Sep. 2011 - Dec. 2017

Harvard University

Bachelor of Arts in Mathematics, Secondary Field Computer Science

Honors: Certificate of Distinction in Teaching (Spring 2008).

Cambridge, MA
Jun. 2009

PUBLICATIONS

When RL Meets Adaptive Speculative Training: A Unified Training-Serving System

J. Wang*, F. Bie*, J. Li, Z. Zhou, Z. Shao, Q. Wu, Y. Liu, Y. Wang, **A. May**, S. Yanamandra, T. Dao, P. Liang, C. Zhang, B. Athiwaratkun, S. Song, C. Xu, X. Wu. ICML 2026

Speculative Speculative Decoding

T. Kumar, T. Dao, **A. May**. ICLR 2026

Minions: Cost-efficient Collaboration Between On-device and Cloud Language Models

A. Narayan*, D. Biderman*, S. Eyuboglu*, **A. May**, S. Linderman, J. Zou, C. Ré. ICML 2025.

MagicDec: Breaking the Latency-Throughput Tradeoff for Long Context Generation with Speculative Decoding

R. Sadhukhan*, J. Chen*, Z. Chen, V. Tiwari, R. Lai, J. Shi, I. Yen, **A. May**, T. Chen, B. Chen. ICLR 2025.

The Mamba in the Llama: Distilling and Accelerating Hybrid Models

J. Wang*, D. Paliotta*, **A. May**, A. Rush, T. Dao. NeurIPS 2024.

SpecExec: Massively Parallel Speculative Decoding for Interactive LLM Inference on Consumer Devices

R. Svirschevski*, **A. May***, Z. Chen*, B. Chen, Z. Jia, M. Ryabinin. NeurIPS 2024.

Sequoia: Scalable, Robust, and Hardware-Aware Speculative Decoding

Z. Chen*, **A. May***, R. Svirschevski*, Y. Huang, M. Ryabinin, Z. Jia, B. Chen. NeurIPS 2024.

Contextual Embeddings: When are they worth it?

S. Arora*, **A. May***, J. Zhang, C. Ré. ACL 2020.

Understanding the Downstream Instability of Word Embeddings.

M. Leszczynski, **A. May**, J. Zhang, S. Wu, C. Aberger, C. Ré. MLSys 2020.

On the Downstream Performance of Compressed Word Embeddings.

A. May, J. Zhang, T. Dao, C. Ré. NeurIPS 2019 (Spotlight, 3% acceptance).

Low-Precision Random Fourier Features for Memory Constrained Kernel Approximation.

J. Zhang*, **A. May***, T. Dao, C. Ré. AISTATS 2019.

Kernel Approximation Methods for Speech Recognition.

A. May, A.B. Garakani, Z. Lu, D. Guo, K. Liu, A. Bellet, L. Fan, M. Collins, D. Hsu, B. Kingsbury, M. Picheny, F. Sha. JMLR 2019.

Compact Kernel Models for Acoustic Modeling via Random Feature Selection.

A. May, M. Collins, D. Hsu, B. Kingsbury. ICASSP 2016.

A Comparison Between Deep Neural Nets and Kernel Acoustic Models for Speech Recognition.

Z. Lu, D. Guo, A.B. Garakani, K. Liu, **A. May**, A. Bellet, L. Fan, M. Collins, B. Kingsbury, M. Picheny, F. Sha. ICASSP 2016.

Filter & follow: How social media foster content curation.

A. May, A. Chaintreau, N. Korula, S. Lattanzi. SIGMETRICS 2014.

SERVICE

Reviewer: NeurIPS, ICLR, ICML, COLM, AISTATS, EMNLP, ICASSP, JMLR. ICML 2019 **best reviewer**, ICASSP 2023 **best reviewer**.